

Responsible Data Analytics Final Report

Delft University of Technology(TU Delft)

Urban Sustainability & Renewable Energy Potential in the Baltic States

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Abstract— We present a four-phase analytics framework to understand and harness renewable energy potential in the Baltic capitals—Vilnius, Tallinn, and Riga—by linking local weather variability to urban sustainability. In the descriptive phase, we map multi-year trends and seasonality in solar radiation, temperature, wind, and humidity, revealing characteristic summer peaks and winter lulls. The diagnostic analysis quantifies interdependencies—e.g. how near-surface humidity and surface temperature jointly modulate clear-sky radiation—which informs when and where renewables perform best. Moving to predictive analytics, we build a time-aware stacking ensemble (random forest, gradient boosting, SVR, MLP with a RidgeCV meta-learner) that, after enforcing a full-year rolling training window, forecasts daily solar irradiance with test RMSEs of 0.58–0.66 kW·h/m²/day ($R^2 \approx 0.93$). Finally, our prescriptive scenarios translate these forecasts into actionable strategies—such as adjusting buffer storage or dynamically curtailing other generators during expected low-insolation spells—to optimize grid integration. Together, these insights equip city planners and utilities with a data-driven playbook for climate-resilient, renewable-rich urban energy systems in northern Europe.

Keywords— Solar Radiation, Data Analytics, Sustainability, Climate Change, Machine Learning, Meteorological Data

INTRODUCTION

As the climate crisis intensifies, renewable energy has become vital to sustainable development. The European Union (EU) aims to reduce CO₂ emissions, boost energy efficiency, and achieve at least 42% renewable energy in its overall mix (Renewable Energy Targets, n.d.)(1). Meeting these goals requires not only innovation but also strategic urban planning, especially in cities, major centers of energy use and emissions.

Urban integration of renewable energy depends heavily on local meteorological conditions, which directly impact the performance of solar, wind, and other weather-dependent systems. This is especially relevant in the Baltic States; Lithuania, Latvia, and Estonia where Vilnius, Riga, and Tallinn are navigating the transition to low-carbon energy while adapting to growing energy demands and climate risks.

Despite access to rich meteorological data, there's limited applied research connecting weather variability to renewable energy output and urban sustainability in this region. This project addresses that gap by analyzing how environmental factors like solar radiation, temperature, wind speed, and

humidity affect renewable energy generation and urban resilience.

Leveraging five years of meteorological data from NASA's POWER platform, the study applies responsible data analytics techniques to uncover both historical trends and predictive insights. The primary objectives are to:

1. Analyze how key weather parameters affect renewable energy potential and urban sustainability in the three capitals.
2. Provide actionable, evidence-based recommendations for energy strategies and urban planning that are tailored to regional climatic conditions and aligned with EU climate policy.

Using a combination of descriptive, diagnostic, and predictive analytics, this project ultimately seeks to translate data-driven insights into prescriptive guidance for policymakers and city planners. The overarching goal is to support sustainable urban transitions and contribute to the broader European effort toward a climate-resilient, low-carbon future.

PROBLEM STATEMENT

To contribute to the EU's renewable energy transition, this project investigates how local meteorological conditions influence renewable energy potential in the Baltic capitals; Vilnius, Tallinn, and Riga. Despite the availability of relevant climate data, there is limited applied research linking weather variability to renewable energy performance and urban sustainability in Vilnius, Tallinn, and Riga. This project addresses that gap by investigating how meteorological parameters affect renewable energy potential in these cities. With our main research question being:

How do variations in meteorological parameters influence renewable energy potential and urban sustainability in Vilnius, Tallinn, and Riga?

To address this question comprehensively, we break it down into four sub-questions aligned with the key phases of data analytics:

1. Descriptive Analytics: What are the historical trends and seasonal patterns of key meteorological variables such as solar radiation, temperature, wind, and humidity?
2. Diagnostic Analytics: How do solar, temperature, moisture, and wind variables correlate and interact, and what implications do these relationships have for renewable energy generation?
3. Predictive Analytics: Can we develop models to predict solar radiation based on other meteorological data that impacts renewable energy generation and potential?

4. Prescriptive Analytics: What general strategies can be used to optimize renewable energy deployment, and what specific actions can be taken based on predicted solar radiation?

By systematically addressing each of these sub-questions, we aim to generate actionable insights that support responsible, climate-conscious energy planning in the Baltic capitals.

Data Description

For this project we utilize five years of meteorological data (2018–2022) from NASA’s POWER platform, focusing on the Baltic capitals: Vilnius, Riga, and Tallinn. These cities were selected due to their increasing importance in the European Union’s green energy transition and their exposure to regional climate variability. The dataset includes key weather parameters related to solar radiation, temperature, humidity, wind, and precipitation, forming the basis for the descriptive, diagnostic, predictive, and prescriptive phases of our analysis.

1.Data preprocessing:

To ensure a clean and consistent foundation for analysis, we began by loading the CSV files. Each dataset was examined to verify its structure and format, ensuring that the same variables were present and consistently labeled across all cities. Once validated, the datasets were merged into a single, unified dataframe to facilitate comparative analysis across locations. We also unified and converted the “YEAR”, “Mo”, and “DY” columns to datetime and Set 'DATE' as the index to have a time series data format to leverage this characteristics of our data. We performed checks for missing values, duplicates, and inconsistencies, confirming that the data was complete and free from redundancies. Data types: All columns were of type float64 (numerical), with the exception of the "City" column, which remains as an object type for categorical identification. This variable will be encoded (e.g., via one-hot encoding) at a later stage during the predictive modeling process. Key preprocessing steps included:

- Standardizing column names (e.g., ensuring uniform capitalization in date fields like "year")
- Checking for missing values: No missing values were found across any rows or columns.
- Duplicate detection: No duplicate entries were present, indicating strong data integrity.

We also performed outlier analysis, see Fig. 1 (The variable WD10M (wind direction at 10 meters) exhibited a broad range of values that visually overwhelmed other variables and since it did not contain any outliers, it is excluded from the plot).

In order to better understand the features that have outliers and to confirm outliers numerically, we double checked them and after assessing they were determined to be genuine observations rather than anomalies caused by data entry errors or noise. As such, these values will be retained during the initial stages of analysis to preserve the integrity and representativeness of the data. If necessary, transformations will be applied in later stages to manage their influence. In particular, when developing machine learning models, techniques such as StandardScaler or RobustScaler from the sklearn.preprocessing module will be considered to minimize

the impact of outliers while maintaining the overall structure and variability of the dataset.

After this process, we also identified potential sources of bias and risk, such as the use of satellite-derived data that may not capture urban microclimate variations accurately. These considerations were noted to ensure a responsible and transparent data analytics process moving forward.

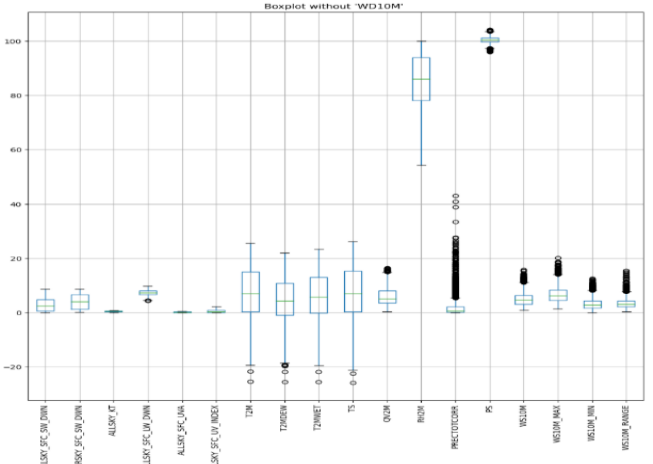


Fig. 1. Box plot of selected meteorological variables (excluding WD10M) for outlier analysis.

Case Sensitivity and Considerations

The focus on Vilnius, Tallinn, and Riga is both timely and socially significant. These cities are under pressure to align with the EU’s 42.5% renewable energy target by 2030, while also navigating challenges related to urban growth, infrastructure modernization, and climate resilience. However, several important limitations must be acknowledged:

- The dataset includes only meteorological variables, meaning it does not reflect socioeconomic, infrastructural, or policy-related factors that are crucial for defining true urban sustainability.
- Insights drawn from just three cities may not generalize across regions with different climates or urban forms, posing a geographic bias.
- The dataset spans only five years, creating a short-term analysis bias and missing longer-term climate trends.
- Urban microclimates (e.g., heat islands) may not be accurately captured by satellite-derived data, raising positionality concerns about treating this data as purely objective.

These limitations highlight the importance of transparency in framing, ensuring that the insights generated are understood within the boundaries of what the dataset can and cannot represent.